











#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

How did we
get here?

The Dual Problem

Primal Problem

$$\min_{\pi \in \mathbb{R}_+^{n \times m}} \langle \mathbf{C}, \pi \rangle$$

$$\text{s.t.} \quad \begin{aligned} \pi \mathbf{1} &= \mathbf{a} \\ \pi^\top \mathbf{1} &= \mathbf{b} \end{aligned}$$

Dual Problem

$$\max_{\mathbf{f} \in \mathbb{R}^n, \mathbf{g} \in \mathbb{R}^m} \mathbf{a}^\top \mathbf{f} + \mathbf{b}^\top \mathbf{g}$$

$$\text{s.t.} \quad f_i + g_j \leq c_{ij}, \quad \forall i, j.$$

Can also be written as: $\mathbf{f} \oplus \mathbf{g} \leq \mathbf{C}$

How did we
get here?

The vectors $\mathbf{f}, \mathbf{g}$ are called “dual potentials” (or Kantorovich, if optimal)

Symmetries:

- $\mathbf{C}$ appears in PP’s objective but DP’ constraint, vice-versa for $\mathbf{a}, \mathbf{b}$
- PP’s has $n \cdot m$ variables, $n + m$ constraints, opposite for DP

Deriving the (Discrete) Dual

Primal Problem: $\min_{\pi \in \mathbb{R}_+^{n \times m}} \langle \mathbf{C}, \pi \rangle \quad \text{s.t.} \quad \pi \mathbf{1} = \mathbf{a}, \pi^\top \mathbf{1} = \mathbf{b}$